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T. N. Weingarten, J. Sprung and A. H. Burgher

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Review

Perioperative management of familial dysautonomia: a systematic review

T. N. Weingarten, J. Sprung, A. H. Burgher

Mayo Clinic College of Medicine, Department of Anesthesiology, Rochester, MN, USA

Summary

Background and objective: Familial dysautonomia (FD), a rare genetic disorder, is characterized by autonomic instability, pulmonary infections, oesophageal dysmotility, spinal abnormalities and episodic 'dysautonomic crisis' characterized by rash, vomiting, sweating and hypertension. Frequent anaesthetic complications have been reported. **Methods:** We performed a comprehensive literature search of perioperative management of FD using an OVID-based search strategy. Identified reports were reviewed to identify perioperative complications as well as anaesthetic techniques and perioperative management strategies developed to minimize or prevent these complications. **Results:** Eighteen case reports or series of perioperative management of FD were identified in the literature for a total of 179 patients undergoing 290 anaesthetics. Intraoperative cardiovascular lability, including cardiac arrests and postoperative pulmonary complications were commonly reported. Preoperative hydration, minimizing the use of volatile anaesthetic agents, postoperative ventilation, use of regional anaesthesia and minimally invasive surgical techniques reduced the incidence of these complications. **Conclusions:** While patients with FD are reported to have a relatively high rate of various perioperative complications, a full understanding of its pathophysiology can be used to develop a perioperative management strategy to anticipate and prevent many of these complications.

Keywords: DYSAUTONOMIA FAMILIAL; HUMAN BEINGS; SURGICAL PROCEDURES; OPERATIVE; ANAESTHESIA GENERAL; SYSTEMATIC REVIEW; INTRAOPERATIVE COMPLICATIONS.

Introduction

Familial dysautonomia (FD), described by Riley and Day in 1949 [1], belongs to the family of disorders known as hereditary and sensory autonomic neuropathies (HSAN). The various types of HSAN are distinguished by mode of inheritance, clinical features, variable degrees of autonomic and sensory nervous system dysfunction and specific genetic abnormalities. FD is classified as HSAN type III using the classification established by Mayo Clinic neurologist Dr Peter Dyck [2]. FD has an autosomal

recessive mode of inheritance with mutations of the gene coding inhibitor of κ light polypeptide gene enhancer in B cells kinase complex-associated protein (IKBKAP) on chromosome 9q31 [3], and most commonly occurs in the Ashkenazi Jewish population. It is differentiated from other HSAN types by profound autonomic dysfunction, hypertension, and orthostatic hypotension, excessive and decreased sweating [2]. Infants present with lack of fungiform papillae and profound dysautonomia with poor suckling, repeated vomiting, failure to thrive, temperature and vasomotor dysregulation associated with hypertension and hypotension [2,4]. Children usually manifest with pulmonary derangements including recurrent aspiration pneumonia, restrictive lung disease from spinal abnormalities, and abnormal responses to hypoxia and hypercarbia

Correspondence to: Toby N. Weingarten, Department of Anesthesiology, Mayo Clinic, 200 First Street SW, Rochester, MN 55905, USA. E-mail: weingarten.toby@mayo.edu; Tel: +507 266 3636; Fax: +507 266 7732

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including apnoea to mild hypoxia [5]. Other clinical features include alacrima, spine disorders including kyphosis and scoliosis and hyporeflexia [2,4]. Emotional stress can trigger a 'dysautonomic crisis' characterized by erythematous macular rash, vomiting, sweating and hypertension. Sensitivity to pain is diminished and patients frequently have corneal abrasions secondary to corneal insensitivity, but this feature is less prominent than in other types of HSAN. Visceral and peritoneal pain sensations are intact.

Anaesthetic management of patients with FD is potentially challenging [6,7]. Haemodynamic instability with anaesthesia has been known since 1959 when Kritchman and colleagues [7] reported two cases of cardiac arrest and six cases of profound hypotension in children undergoing general anaesthesia. Atelectasis was reported in 17% of children who underwent gastrointestinal surgical procedures [6]. Postoperative intractable vomiting has also been reported [6]. Because this is a very rare disorder, large-scale studies are not feasible and the knowledge regarding anaesthetic management emerges from case series and reports. This report is a systematic review of the literature to summarize the existing experience of perioperative management of patients afflicted with this disorder.

Methods

We performed a comprehensive literature search using an OVID-based search strategy of (1) MEDLINE (1966–present); (2) EMBASE (1988–present); (3) Current contents (1966–present); and (4) Scopus (1996–present) under the following keywords: familial dysautonomia, Riley-Day syndrome, hereditary sensory and autonomic neuropathy type III, and an(a)esthesia, all limited to human beings and the English language. Bibliographies of identified published reports were searched for additional references. Identified reports were reviewed to identify perioperative complications unique to this patient population as well as anaesthetic techniques and perioperative management strategies developed to minimize or prevent these complications.

Results

The OVID-based search strategy identified 16 case reports or case series of perioperative management of patients with FD (see Table 1 for a summary of perioperative management and complications). Review of the bibliographies of the manuscripts identified two other reports. A case report written in German with only the abstract translated into English was excluded from this review, but this case did report that the patient experienced intraoperative

and postoperative respiratory difficulties requiring prolonged postoperative ventilation, haemodynamic lability and autonomic crises [8].

A total of 179 patients underwent 290 anaesthetics. Eight of the reports present the anaesthetic experience of only a single patient, but six of these eight do report multiple episodes of anaesthetic care in the same patient [9–15]. Intraoperative complications were primarily cardiovascular with haemodynamic variability including episodes of hypotension, hypertension, bradycardia and dysrhythmias [6,7,9–13,16–19], complete heart block on one occasion [9] and three occurrences of cardiac arrest that were successfully treated [7,13]. Intraoperative oxyhaemoglobin desaturation was also reported on occasion [6]. Postoperative complications included primarily cyclic vomiting and pulmonary complications such as pneumonia, atelectasis, excessive secretions and respiratory arrest requiring reintubation on two occasions [6,7,9,12,16,19–21]. Haemodynamic lability was reported in the postoperative period [6,7,9,12,16,19–21], but less frequently than in the intraoperative period [6,7,9–13,16–19]. One patient experienced a postoperative seizure [9] and one patient intraoperative confusion [15].

Perioperative management of these patients evolved with time, reportedly in an effort to mitigate these frequent complications. Early management strategies consisted primarily of premedication followed by various general anaesthetic techniques that consisted of nitrous oxide alone, nitrous oxide supplemented with various anaesthetic agents or anaesthetic agents alone [7,9–11,16]. In early reports, now obsolete anaesthetics such as tribromoethanol and the volatile agents ether, methoxyflurane and cyclopropane were used [7,9,16]. The two cardiac arrests reported by Kritchman and colleagues [7], developed during anaesthetics maintained by nitrous oxide, supplemented by thiopental sodium in one case and by tribromoethanol in the other case. The cardiac arrest reported by Kaufman and colleagues [13], was associated with the use of an unspecified volatile anaesthetic. McCaughey and colleagues [9] demonstrated that administration of methoxyflurane, ether or halothane resulted in hypotension and bradycardia, and haemodynamics were more stable with nitrous oxide alone. Postoperative care was not reported in these early series.

Beilin and colleagues [18] reported aggressive preoperative management including intravenous (i.v.) hydration with fresh frozen plasma; pulmonary optimization with chest physiotherapy, deep suctioning, antibiotics and tracheostomy in one patient; and administration of cimetidine. Anaesthesia

Table 1. Summary of perioperative management and complications of reported cases of familial dysautonomia.

Reference and year	n patients/ n cases	Procedural type	Pre-surgical care	Premedication	Induction	Anesthesia maintenance	Intraoperative complications	Post-surgical care	Post-surgical complications
Kritchman (1959) [7]	7/27	3 – Otolaryngology 18 – Ophthalmology 2 – Orthopedic 5 – Neurosurgery	NR	Belladonna – 24 Opiate – 9 Barbiturate – 7 Chlorpromazine – 2 None – 3	NR	Tribromoethanol – 3 Tribromoethanol/ Ether – 5 Vinyl ether/Ether – 2 Cyclopropane/Ether – 2 Ethyl chloride/Ether – 2 Ether – 6 STP/N ₂ O – 4 Tribromoethanol/ STP/N ₂ O – 1 STP – 1 Cyclopropane – 1	Hypotension – 6 Cardiac arrest – 2	NR	Pneumonia – 5 Atelectasis – 2 Cyanosis – 2 Cyclic vomiting – 3 Hypotension – 1 Fever – 1
McCaughey (1965) [9]	1/6	Spine	Anxiolysis	Atropine – 5 Chlorpromazine – 2 Meperidine – 5	STP	N ₂ O/ Methoxyfluorane – 2 N ₂ O – 1 Hal/Methoxyflurane – 1 Methoxyflurane – 1 Ether – 1	Hypotension – 6 Complete heart block – 1		Vomiting Hypertension Seizures Fever
Bartels (1970) [10]	1/2	1 – Otolaryngology 1 – Thoracic	NR	Chlorpromazine Meperidine	N ₂ O	N ₂ O	Hypotension	NR	None
Inkster (1971) [11]	1/1	Laparotomy	NR	Trimeprazine Atropine Propranolol	N ₂ O/Hal	N ₂ O/Hal	Bradycardia	NR	None
Meridy (1971) [16]	8/24	3 – Ophthalmology 8 – Otolaryngology 6 – Laparotomy 6 – Orthopedic	NR	Chlorpromazine – 6 Atropine – 19	STP – 9 Methohexitone – 3 N ₂ O/Hal – 5 Ethyl chloride – 4	Ether – 6 N ₂ O/Hal – 6 N ₂ O/Hal/ Methoxyflurane – 4 N ₂ O – 5	Hypertension Hypotension	NR	Hypertension Hypotension Lung infection Aspiration Pneumonitis Copious secretions Re-intubation – 2 Vomiting Fever Decubitus Ulcer – 1 Cyclic vomiting – 2
	NR	1 –Plastics			Ether – 2 N ₂ O – 1	Ether/Hal – 1 N ₂ O/Methoxyflurane – 2			
Stenqvist (1982) [12]	1/2	Spine	NR	Atropine Promethazine	Enflur/N ₂ O – 2	Enflur – 2	Hypotension Hypothermia	NR	
Kaufman (1982) [13]	1/multiple	Dental	NR	NR	NR	GA – 1 N ₂ O – others	Cardiac arrest	NR	NR
Cox (1983) [17]	2/2	Laparotomy	NR	Diazepam – 1	STP/N ₂ O/Hal – 1	N ₂ O/Hal – 2	Labile blood pressure	NR	None
Foster (1983) [39]	2/2	Spine	NR	None – 1 NR	N ₂ O/Hal – 2 NR	N ₂ O	NR	NR	NR
Freeman 1983 [14]	1/2	Orthopedic	None	Chlorpromazine	N ₂ O	N ₂ O	None	None	None

Table 1. *Continued*

Reference and year	n patients/ n cases	Procedural type	Pre-surgical care	Premedication	Induction	Anesthesia maintenance	Intraoperative complications	Post-surgical care	Post-surgical complications
Beilin (1985) [18]	4/6	4 – Laparotomy 2 – Otolaryngology	Chest PT Deep suction Antibiotics Cimetidine IV hydration – FFP	Diazepam – 3 Flunitazepam – 5	Diazepam and Flunitazepam	Fentanyl/N ₂ O	Hypertension – 1 Mild hypotension with induction – 1	Post-operative ventilation	None
Axelrod (1988) [6]	81/127	58 – Laparotomy 18 – Otolaryngology 13 – Genitourinary 11 – Ophthalmology 13 – Orthopedic 6 – Neurosurgery 5 – OB/GYN 2 – Thoracic 1 – Dental OB/GYN	IV hydration Anxiolysis Cimetidine>1979	Chlorpromazine<1975 Diazepam>1975	STP	N ₂ O/Hal N ₂ O/Enflur Fentanyl/GA>1986	Hypotension – 4 Oxyhaemoglobin desaturation – 2	Post-operative ventilation Chest PT Anxiolysis IV hydration	Hypotension – 1 Cyclic vomiting – 16 Pneumonia – 8 Fever – 6 Atelectasis – 10
Leiberman (1991) [15]	1/1		NR	NR	None	Infiltration with 1% lidocaine	Intraoperative confusion	None	None
Udassin (1992) [20]	34/34	Laparotomy	Chest PT IV hydration – crystalloid & FFP	NR	NR	Fentanyl/GA	NR	Post-operative ventilation Chest PT	Hypotension Tachycardia
Szold (1996) [21]	26/26	13 – Laparotomy	1–2 days pre-op chest PT	NR	NR	Fentanyl/GA	NR	Laparotomies – post-operative ventilation	Atelectasis Laparotomies – 9 Pulmonary complications
Dell'oste (1996) [22]	1/16	13 – Laparoscopic procedures 5 – Endoscopy 1 – CT scan 2 – Orthopedic 7 – Dental 1 – Ophthalmology	IV hydration IV hydration	Atropine – 16 Diazepam – 9 Meperidine – 5	Ketamine – 3 STP – 11 Althesin – 2	Ketamine – 4 Hal – 3 Iso – 3 Enflur – 2 N ₂ O – 13	None with ketamine	NR	None with ketamine
Challands (1998) [19]	3/6	Laparotomy	NR	Chloral Hydrate – 1 Temexepam – 3 Diazepam – 1 Atropine – 1 Midazolam	Cyclopropane – 2 N ₂ O/Hal – 1 Barbiturate – 2	N ₂ O/Iso/Fent – 2 N ₂ O/Iso + Epidural – 3 N ₂ O/Hal + Epidural – 1	Cardiovascular lability – 2 None with epidural	Epidural catheter for 2–3 days – 4	Several crises – 1 Desaturation – 1 None with epidural
Wengrower (2002) [23]	4/4	Endoscopy	IV hydration Anxiolysis		Propofol	Propofol	None	None	None

Ether: diethyl ether; Hal: halothane; Enflur: Enflurane; Iso: Isoflurane; N₂O: nitrous oxide; STP: sodium thiopental; i.v.: intravenous; FFP: fresh frozen plasma; NR: not reported; GA: general anaesthetic; PT: physiotherapy; OB/GYN: obstetrics/gynecological; '-': number of cases or number of treatments; '<': before; '>': after.

was induced with a benzodiazepine and fentanyl and maintained with nitrous oxide with fentanyl supplementation and intraoperative haemodynamic perturbations were less frequent and milder than in the previous series [18]. Postoperative care consisted of ventilation for several hours and there were no reported postoperative pulmonary complications [18]. Axelrod and colleagues [6] reported preoperative care that consisted of i.v. hydration, anxiolysis and cimetidine. Initially, patients' tracheas were extubated in the recovery room and post-laparotomy atelectasis was reported as often as 57% [6]. In the second part of their series, patients remained intubated for as long as 72 h after the procedure, received aggressive chest physiotherapy and broad spectrum antibiotics, and post-laparotomy atelectasis complicated only 12% of cases [6]. However, the criteria used to make the diagnosis of atelectasis in the Axelrod series was not offered, introducing a degree of ambiguity. Dell'oste and colleagues [22] reported good outcomes with a ketamine anaesthetic technique. Challands and colleagues [19] reported reduced postoperative complications with placement of an epidural catheter for postoperative analgesia in patients undergoing Nissen funduplications. Leiberman and colleagues [15] reported a successful caesarean section performed with only local infiltration of 1% lidocaine. Most recently, Wengrower and colleagues [23] reported successful management of endoscopic procedures under heavy sedation with i.v. midazolam and propofol. Patients were reported to respond normally to both depolarizing and non-depolarizing muscle relaxants [6,10,16–19,22]. Besides cimetidine used to neutralize gastric secretions and chlorpromazine used as a sedative, prophylactic anti-emetics were not reportedly administered.

Discussion

Cardiovascular and haemodynamic effects

Patients with FD suffer derangements of several organ systems that present challenges to the anaesthesiologist. Haemodynamic instability is one of the hallmarks of the disorder. Orthostatic hypotension without compensatory tachycardia [24,25] is common. There is a congenital reduction in the number of peripheral sympathetic neurons in the sympathetic ganglia but further degeneration continues with age [26,27]. There is a lack of sympathetic innervation of blood vessels [28]. The parasympathetic nervous system and the baroreflex are dysfunctional and this may explain the abnormal cardiac response to changes in blood pressure (BP)

[25,29]. Patients are often chronically dehydrated with reduced intravascular volume which contributes to haemodynamic instability. Catecholamine metabolism is abnormal in FD [30]. Plasma levels of the enzyme that converts dopamine to norepinephrine, dopamine β -hydroxylase, is reduced in these patients [31]. Orthostatic hypotension can be particularly problematic, and patients are encouraged to maintain adequate hydration, a high salt diet and may be medicated with mineralocorticoids, such as fludrocortisone and midodrine [32]. Patients experience paroxysmal dysautonomic crises, often triggered by emotional upset, characterized by vomiting, rash, sweating as well as hypertension [33]. It has been hypothesized that the observed hypertension is secondary to an increased sensitivity of the vasculature to circulating catecholamines released during a crisis. Exaggerated response by skin vessels in patients with FD to norepinephrine has been observed [34]. These patients also have a greater increase in BP in response to systemic administration of norepinephrine compared to healthy controls [35]. General anaesthetics with volatile agents were noted to precipitate haemodynamic instability [9,12] and may be associated with cardiac arrest [13]. Isoflurane was reported to be associated with haemodynamic instability [19]. Efforts to supplement volatile anaesthetics with i.v. fentanyl or regional anaesthesia [6,18,19] or with i.v. anaesthetic techniques [22,23] seem to reduce these negative effects. It is difficult to elucidate retrospectively the primary role of inhalational anaesthetics such as ether, methoxyflurane etc., in the pathogenesis of some of the problems described with these patients. More specifically, at that time, four to five decades ago, lack of both contemporary monitoring and/or today's standards of airway management could have been a contributing factor. Several decades ago anaesthetic overdose and problems with ventilation were frequent causes of anaesthesia attributed perioperative morbidity and mortality in the general surgical population [36]. Assessment of effects of an easily titratable volatile agent such as sevoflurane or desflurane has not been reported to date in dysautonomic patients.

State of hydration

Patients with FD may suffer from chronic dehydration secondary to dysphagia [6]. Intraoperative hypotension in these patients was attributed to inadequate hydration or blood loss and as all patients favorably responded to infusion of blood or crystalloid and reduction of inhaled anaesthetic concentrations [6]. In several reports, preoperative

hydration was useful to optimize haemodynamics [6,18,20–23]. Some authors conducted preoperative hydration with fresh frozen plasma, but they did not provide rationale or evidence that plasma is superior over other safer alternative fluids such as crystalloids and colloids [18,20].

Respiratory effects

Pulmonary derangements are common secondary to frequent aspiration that predispose patients to pneumonia and bronchiectasis [2]. The kyphosis frequently encountered in these patients can cause restrictive lung disease [4]. Hypotonia [2] can diminish postoperative pulmonary cough strength. Furthermore, FD patients demonstrate an abnormal response to moderate hypoxia with development of hypoventilation, hypotension and bradycardia [5]. Aggressive preoperative optimization of the pulmonary status is prudent if the patient shows signs of pulmonary congestion [18]. Improved outcome with prolonged tracheal intubation following laparotomy has been demonstrated [6], but pulmonary complications were still described [21]. Minimally invasive techniques such as laparoscopic or endoscopic surgery are associated with fewer postoperative pulmonary complications [21,23]. Postoperative epidural analgesia after open laparotomy was also shown to be beneficial [19].

Gastrointestinal disorders

Gastrointestinal manifestations of the syndrome include abnormal motility of the esophagus resulting in gastroesophageal reflux [2]. Patients also suffer dysautonomic crises characterized by paroxysms of vomiting, hypertension, tachycardia and personality changes [2]. Functional neuroimaging of FD patients revealed that dysautonomic crises are associated with abnormal cerebral perfusion in the temporal and frontal lobes, similar to abnormalities seen in patients with seizure activity [37]. The observation that the administration of benzodiazepines is effective in treating dysautonomic crises further supports the hypothesis that these episodes arise from cerebral dysfunction. Clonidine has also been used in conjunction with benzodiazepines to control associated hypertension during crises. These manifestations may result in aspiration pneumonia, malnutrition, chronic dehydration and electrolyte abnormalities. Patients may be on proton pump inhibitors or H₂-antagonists as management of gastroesophageal reflux. Fundoplication has been demonstrated to mitigate many of these symptoms [20]. Postoperative vomiting was frequently reported and treated with diazepam

[6], promethazin [12] and dimenhydrinate [9]. The effects of potent 5-HT₃ antagonists on postoperative nausea and vomiting in FD have not been reported. Rapid sequence induction with application of cricoid pressure was recommended to reduce the risk of aspiration with endotracheal intubation [6]. However, others did not perform rapid sequence induction [18], or secure the trachea [22,23] and did not encounter complications from pulmonary aspiration.

Neurologic and pain effects

Mental retardation can be present in patients with FD. Emotional upset can trigger a dysautonomic crisis characterized by hypertension, erythematous macular skin rash, excessive sweating and intractable vomiting, making preoperative anxiolysis an important priority. Attention must also be paid to protecting the eyes of FD patients as they suffer from lacrimal secretory disorders (alacrima) and corneal insensitivity [2]. However, corneal abrasions were not reported in the reviewed cases. Hypo-sensitivity to pain is a feature of FD; however, opiate analgesics administration was reported in most reports, and regional anaesthetic techniques were associated with favorable outcomes [19]. This is in contrast to the anaesthetic management of patients with other HSAN disorders who frequently do not require any perioperative analgesics and typically tolerate general anaesthetics without major complications [38]. As with all HSAN disorders, temperature dysregulation can manifest in FD [2]. Postoperative fever was reported, but it was unclear if this was secondary to more common causes of postoperative fevers such as infection or atelectasis, or due to primary temperature dysregulation. Of note, there is no association between malignant hyperthermia and FD, as the genetic mechanisms of precipitating hyperpyrexia in malignant hyperpyrexia and FD are fundamentally different.

In conclusion, FD has profound implications for general anaesthesia. While these patients were reported to have a relatively high rate of various perioperative complications, these reports date to the time of obsolete inhalational anaesthetics. The safety of newer volatile anaesthetic agents, such as sevoflurane and desflurane, has not been assessed in these patients. Use of minimally invasive surgical techniques, regional anaesthesia, and i.v. techniques appears to contribute to reduction of the complication rate. Correction of chronic dehydration with preoperative hydration seems to reduce intraoperative haemodynamic instability. Preoperative optimization of pulmonary status by treating underlying infections and aggressive respiratory

therapy is also prudent. Aggressive postoperative pulmonary management can further decrease respiratory complications. I.v. fluid boluses often correct hypotension and vasopressors should be used with caution as patients may be hypersensitive to these agents. Vomiting often responds to benzodiazepines. The perioperative management of patients with FD requires full understanding of its pathophysiology in order to anticipate and prevent possible complications.

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